

EC-Earth V2.2: description and validation of a new seamless earth system prediction model

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Abstract EC-Earth, a new Earth system model based on the operational seasonal forecast system of the European Centre for Medium-Range Weather Forecasts (ECMWF), is presented. The performance of version 2.2 (V2.2) of the model is compared to observations, reanalysis data and other coupled atmosphere–ocean–sea ice models. The large-scale physical characteristics of the atmosphere, ocean and sea ice are well simulated. When compared to other coupled models with similar complexity, the model performs well in simulating tropospheric fields and dynamic variables, and

performs less in simulating surface temperature and fluxes. The surface temperatures are too cold, with the exception of the Southern Ocean region and parts of the Northern Hemisphere extratropics. The main patterns of interannual climate variability are well represented. Experiments with enhanced CO₂ concentrations show well-known responses of Arctic amplification, land-sea contrasts, tropospheric warming and stratospheric cooling. The global climate sensitivity of the current version of EC-Earth is slightly less than 1 K/(W m^{−2}). An intensification of the hydrological cycle is found and strong regional changes in precipitation, affecting monsoon characteristics. The results show that a coupled model based on an operational seasonal prediction system can be used for climate studies, supporting emerging seamless prediction strategies.

This paper is a contribution to the special issue on EC-Earth, a global climate and earth system model based on the seasonal forecast system of the European Centre for Medium-Range Weather Forecasts, and developed by the international EC-Earth consortium. This special issue is coordinated by Wilco Hazeleger (chair of the EC-Earth consortium) and Richard Bintanja.

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1 Introduction

Climate and weather forecasting applications share a common ancestry and also build on the same physical principles. Nevertheless, climate research and weather forecasting are seen as different applications. The concept of “seamless prediction” is emerging to forge weather forecasting and climate change studies into a joint topic (Palmer et al. 2008). Another development is the extension of the physical climate system towards a comprehensive view of the Earth system by including interactions with the biogeophysical components. Here, we report on the newly developed EC-Earth model, which merges these developments. The general strategy has been laid out by Hazeleger et al. (2010), here a more detailed description and evaluation of the model system is presented.

Modelling strategies have been developed to deal with different types of uncertainties that are relevant to climate

predictions and projections. Initial condition errors are relevant in seasonal and decadal predictions using initial states estimated from observations (e.g. Palmer et al. 2004). Model errors are covered with either multi-model ensembles (Meehl et al. 2007, hereafter CMIP3), perturbed physics ensembles (Murphy et al. 2004) or stochastic parameterizations (e.g. Shutts et al. 2008). The multi-model ensemble of CMIP3 (Meehl et al. 2007) also samples boundary condition uncertainty by prescribing a range of emissions and concentrations of greenhouse gases and aerosols. New developments are the inclusion of the biosphere in models in order to simulate for instance the carbon cycle. With EC-Earth contributions to these different types of model experimentation are foreseen. For instance, as part of the Coupled Model Intercomparison Project Phase 5 (CMIP5, Taylor et al. 2011) initialized decadal hindcasts and forecasts are made, as well as climate projections with different concentration scenarios.

The numerical weather prediction (NWP) system of the European Centre for Medium-Range Weather Forecasts (ECMWF) forms the basis for EC-Earth. NWP models are designed to capture the atmospheric fluctuations and fast feedbacks accurately. In addition, NWP models are applied at daily to seasonal time scales and include data assimilation capabilities. Best practices from the NWP community can be taken up in climate science when applicable. Vice versa, elements developed in Earth System modelling can be adopted in the NWP models. The EC-Earth version described here differs slightly from the ECMWF operational model cycle. The differences with the ECMWF model imply that the seamless prediction strategy applied here is not very strict. However, the EC-Earth version that is used here is applied across a range of time scales. It is used for initialized seasonal to decadal predictions and for climate projections. These issues are explained in more detail in Hazeleger et al. (2010).

In this paper we describe a version of EC-Earth similar to the configuration that will be used for the CMIP5 project. An atmosphere-only version of the system has been used in a number of studies (Haarsma et al. 2009; Storelvmo et al. 2009; Van der Molen et al. 2011) and a coupled version is used by Bintanja et al. (2011). The aim of this paper is to describe the components of the coupled atmosphere–ocean–sea ice–land model. We evaluate the present-day climate from a long coupled simulation. Different aspects of the mean climate and its variability will be highlighted. In addition, we will investigate the near-equilibrium response to greenhouse gas changes using simulations with altered greenhouse gas concentrations. The version of EC-Earth used for CMIP5 will differ in the treatment of aerosols. Except for different prescribed aerosol concentration fields, that version will include a dependency of aerosol optical properties on atmospheric

humidity and a rescaled inhomogeneity factor (see below). The differences between the control climates of both model versions are small. The version used for CMIP5 is somewhat colder which is particularly reflected in more sea ice. Forthcoming publications on the CMIP5 results with EC-Earth will clarify these differences.

2 Model description

EC-Earth V2.2 is based on ECMWF's seasonal forecasting system 3 (<http://www.ecmwf.int/products/forecasts/seasonal/documentation/system3/index.html>). Here we describe the various components of the coupled model. In contrast to the system used by ECMWF, an interactive atmosphere–ocean–sea ice coupling is applied across the entire globe, including the polar regions. Different sea ice and land modules are used. The ocean wave model in use by ECMWF is not used in our application. However, the main difference with the model system of ECMWF is the coupling to another ocean model, NEMO version 2, described below. Although EC-Earth includes a coupling to the atmospheric chemistry and aerosol model TM5 (Huijnen et al. 2010), no interactive coupling is used here. In this manuscript we will focus on the atmospheric physics, dynamics, ocean, sea ice and land component of EC-Earth. The model runs presented do not employ any flux correction or anomaly coupling.

2.1 The atmosphere model

The atmosphere model is the Integrated Forecasting System (IFS) cycle 31r1. This is a primitive equation model with fully interactive cloud and radiation physics. A detailed description of the dynamics and the physics of the model can be found at ECMWF: <http://www.ecmwf.int/research/ifsdocs/CY31r1/index.html>. The model can be run at different resolutions. Here, we use a horizontal spectral resolution of T159 (triangular truncation at wavenumber 159). The computation of physical processes is done on the corresponding to 1.125°, reduced Gaussian grid N80, except for the radiation scheme which uses a lower resolution grid (the reduced Gaussian grid associated with the spectral truncation T63). There are 62 layers in the vertical up to 5 hPa. IFS employs a semi-Lagrangian advection scheme with a time step of 1-hour, which makes the model very efficient. Here we describe the changes with respect to IFS cycle 31r1. For further details we refer to the documentation at ECMWF.

A small number of changes to the physics parameterizations have been made to optimize the model for climate simulations without jeopardizing the prediction skill at short time scales. First, an improved description of the

entrainment of environmental air in deep convecting plumes, that was introduced the more recent cycle 32r3, has been adopted. This new convection scheme greatly improved precipitation patterns over the tropics and circulation characteristics in the midlatitudes (Bechtold et al. 2008; see also the documentation of IFS c33r1 <http://www.ecmwf.int/research/ifsdocs/CY33r1/index.html>). Finally, the inhomogeneity scaling factor for shortwave cloud optical thickness has been reduced from 0.7 to 0.57 and an improved mass conservation correction scheme has been applied (the scheme from IFS cy33R2). These changes led to a significantly better mean atmospheric state.

In contrast to ECMWF's cycle31r1, EC-Earth optionally allows for time varying aerosol forcings based on historical and projected concentrations (Lamarque et al. 2011). These will be applied in the CMIP5 simulations. For the simulations presented here, a dataset of aerosol concentrations based on the aerosol model intercomparison carried out for the IPCC Third Assessment Report was used (Penner et al. 2001). The same dataset was used in a more recent model intercomparison of indirect aerosol effects (Penner et al. 2006). Present-day concentrations in this dataset correspond to approximately the year 2000. In the pre-industrial simulation only the natural contributions are included. Although indirect aerosol forcing has been implemented in EC-Earth, we chose not to use it here because different parameterizations yield highly variable results (see Storelvmo et al. 2009).

2.2 Land surface component

The land surface component in IFS cycle 31r1 is modelled using the TESSEL (Tiled ECMWF Scheme for Surface Exchanges over Land) scheme (van den Hurk et al. 2000). In EC-Earth we have replaced it by H-TESSEL, which uses an improved representation of hydrology, as in more recent cycles of IFS (Balsamo et al. 2009). The updates comprise an explicit parameterization of fast surface runoff, the use of spatially varying soil textures, a different formulation of the soil hydraulic properties (Van den Hurk and Viterbo 2003), and a revision of the snow density and albedo formulation (Dutra et al. 2010). The model has 4 active soil layers extending to 2.89 m depth, without considering capillary rise of groundwater or horizontal exchange of soil water.

Soil water flow is described by the diffusivity form of the Richards equation using the same 4-layer mesh as for soil temperature. The hydraulic conductivity and diffusivity are described with the analytical functions proposed by van Genuchten (1980). Runoff is collected in pre-defined river basins and distributed instantaneously into ocean points nearby the outlets of major rivers, without considering delayed river transport.

2.3 Ocean model

EC-Earth contains the NEMO Version 2 ocean model (Madec 2008), which consists of the Océan Parallélisé version 9 (OPA9) ocean general circulation model and the Louvain-la-Neuve Ice Model version 2 (LIM2) sea ice model (see next subsection). OPA9 is a primitive equation model with a free surface, and an energy- and enstrophy-conserving form of the momentum advection. The lateral boundary condition on the momentum equations is free slip for the global configuration. The horizontal momentum diffusion uses a combination of harmonic and biharmonic operators. The tracer equations use a TVD advection scheme (Zalesak 1979). The diffusion operator of temperature and salinity is Laplacian and operates to diffuse isopycnal thickness according to the adiabatic mixing scheme of Gent and McWilliams (1990). The vertical coordinate system is based on fixed geopotential levels with partial cell thicknesses allowed at the sea floor. The vertical mixing uses the Turbulent Kinetic Energy scheme of Gaspar et al. (1990) (see also Delecluse and Madec 2000). This is a single-equation scheme with an algebraic expression for the mixing length based on the local density profile. A quadratic bottom friction boundary condition is applied. We use the ORCA-1 configuration, which has a C-grid and about 1° horizontal resolution and 42 layers. The model has higher resolution of about one third of a degree near the equator to resolve equatorial planetary waves. The grid has three poles; one on the South Pole and two near the North Pole (one in Canada and one in Siberia), but at different longitudes. As a result, the horizontal resolution is increased in the vicinity of the North Pole. More detail can be found in Sterl et al. (submitted to this special issue).

2.4 Sea ice model

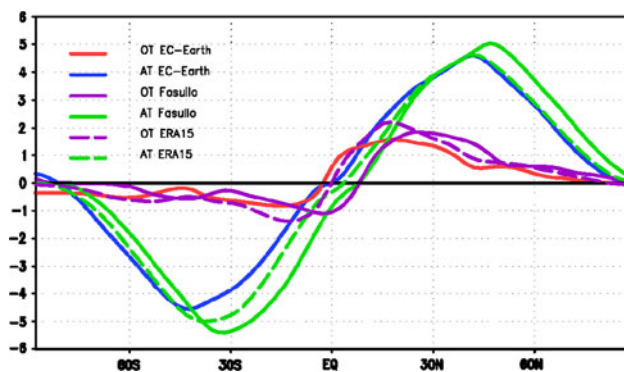
The sea ice component is currently modelled using the 2nd version of the Louvain-la-Neuve (LIM2) model (Fichefet and Morales Maqueda 1997). This has viscous-plastic dynamics (Hibler 1979) and a 3-layer thermodynamic model (Semtner 1976).

2.5 Coupler

The IFS/H-TESSEL and NEMO/LIM2 are coupled with the Ocean Atmosphere Sea Ice Soil version 3 (OASIS3; Valcke 2006) coupler. NEMO receives surface heat flux, snowfall, freshwater flux, and wind stress from IFS via the OASIS3 coupler. The heat flux is separated into a solar and non-solar part, and calculated separately over water and sea ice. Net freshwater flux (precipitation minus evaporation) includes runoff as provided by H-TESSEL. Glacier calving

Table 1 Top of the atmosphere and surface energy fluxes in W m^{-2}

	EC-Earth	Kiehl and Trenberth (1997)	Trenberth et al. (2009)	ERA-Interim
Top net shortwave	242.6	235	239.4	244.0
Top net long wave	−242.7	−235	−238.5	−245.5
Net top	−0.1	0	0.9	−1.5
Surface net shortwave	163.1	168	161.2	163.9
Surface shortwave down	186.4	198	184	187.7
Surface shortwave up	−23.3	−30	−23	−23.8
Surface net long wave	−61.5	−66	−63	−56.3
Surface longwave down	335.8	324	333	341.2
Surface long wave up	−397.3	390	−396	−397.5
Sensible heat flux	−18.3	−24	−17	−17.5
Latent heat flux	−82.7	−78	−80	−82.8
Net surface	0.6	0	0.9	7.9

**Fig. 1** Meridional heat transport (10^{15} W) in the atmosphere (AT) and ocean (OT). Results derived from the EC-Earth PD run are compared with ERA15 reanalysis data from Trenberth and Caron (2001) and estimates from Fasullo and Trenberth (2008)

is included by spreading accumulation of snow over a threshold of $10,000 \text{ kg m}^{-2}$ over nearby ocean points in the summer season, in case of a warming net surface heat flux. Sea surface temperature, surface ocean velocity, sea ice fraction, sea ice temperature, and snow thickness over sea ice are transferred from NEMO to IFS/H-TESSEL via the coupler. These physical fields are used for the calculation of heat, freshwater, and momentum fluxes in the atmosphere. The coupling frequency between the ocean and atmosphere is every 3 h.

3 Experimental setup

Three long control runs are analyzed in this paper. All runs use the same vegetation and albedo maps that are used in IFS cy31r1, representing present-day conditions. Greenhouse gas concentration in the atmosphere, most notably carbon dioxide, and the solar constant are imposed as boundary conditions to the models and are different for the three runs.

The first run (labelled PI) uses pre-industrial boundary conditions. That is, the CO_2 concentration in the atmosphere has been set to 285 ppm. The solar constant is the average of the estimated strength between the years 1844 and 1856, which varied between 1366.2 and 1365.6 W m^{-2} . We initialised the model from temperatures and salinities from the World Ocean Atlas 2001 (Conkright et al. 2002) in the ocean, with initial ocean currents at rest. The initial atmospheric state was taken from ERA-Interim reanalysis fields for the year 1990. The run is 730 years long, of which the last 300 years are analyzed in this paper.

A second run was initialized from the end of the pre-industrial simulation. It uses perpetual boundary conditions from the year 2000. This present-day (PD) run uses a CO_2 concentration of 367 ppm. The solar forcing is the average from 1996–2008, which varied between 1366.7 and 1365.6 W m^{-2} . This simulation has been run for 440 years. We use the last 300 years in this paper.

Finally, in a third run the CO_2 was quadrupled instantaneously with respect to the PD run. This run was initialized from the end of the PD simulation and has been run for 176 years. We use the last 132 years, acknowledging that this run still contains substantial drift in the net radiation components and the deep ocean. The top of the atmosphere net imbalance is 1.4 W m^{-2} which indicates a warming trend. However, the tropospheric trends are strongly reduced after 4 decades.

We analyze the mean state, seasonal cycle and inter-annual variability of the model as well as the response to increased carbon dioxide concentrations. For the atmosphere, we compare the model results to both the ERA-40 (January 1959–January 2001; Uppala et al. 2005) and ERA-Interim data (January 1989–January 2009, for the radiation balance we use data from January 1979–January 2010; Dee et al. 2011). For precipitation we use the CPC Merged Analysis of Precipitation (CMAP) data based on

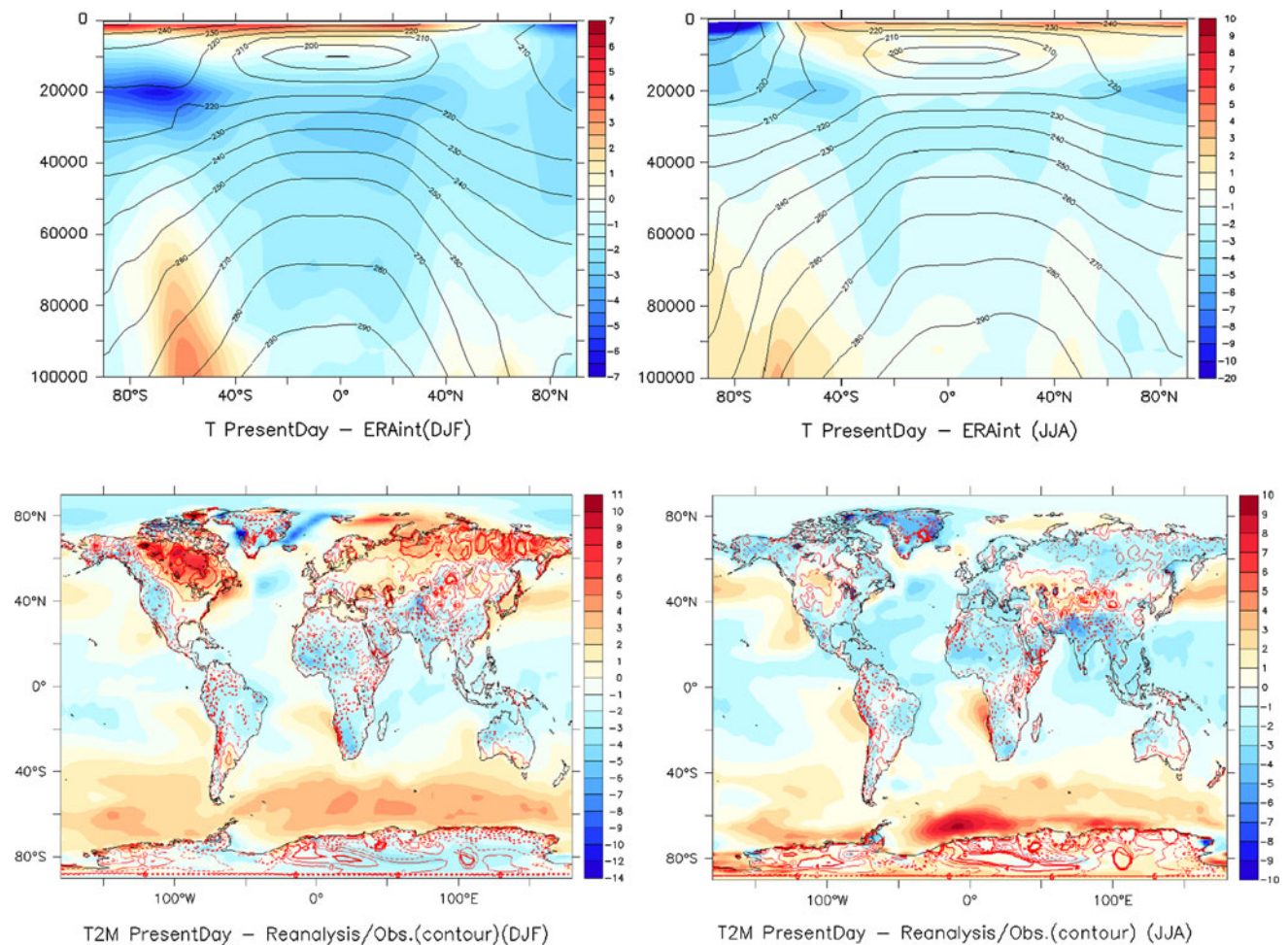


Fig. 2 Top Zonal mean temperature in K (contours) in the EC-Earth PD simulation and the difference with ERA-Interim data (colours). Bottom 2-m temperature difference between the EC-Earth PD

simulation and the ERA-Interim data (colours) and observational estimates of Fan and van den Dool (2008) (contours). Left average over December, January, February. Right average over June, July, August

Table 2 Performance indicators according to the metric of Reichler and Kim (2008)

EXP	T2m	MSL	Pr	Q_n	τ_x	τ_y	T	u	v	q	SST	Ice	SSS	Total (1 = mean of CMIP3)
Mean CMIP3	25.1	11.7	38.9	14.2	4.0	3.1	39.0	12.1	8.3	29.4	17.2	0.3	0.22	1.0
Best CMIP3	16.7	2.8	15.1	12.5	2.7	2.3	41.7	7.1	4.6	19.5	8.5	0.2	0.1	0.64
Worst CMIP3	97.6	5.6	41.0	13.0	4.8	4.0	46.8	8.6	10.2	30.3	18.5	1.0	0.1	1.30
EC-Earth	32.4	1.5	36.0	22.8	7.4	4.4	23.9	2.2	6.9	12.7	19.5	0.1	0.1	0.85

T2m 2-meter temperature, *MSL* mean sea level pressure, *Pr* total precipitation, Q_n net surface heat flux, τ_x zonal surface wind stress, τ_y meridional surface wind stress, *T* atmospheric temperature, *u* atmospheric zonal wind, *v* atmospheric meridional wind, *q* atmospheric humidity, *SST* sea surface temperature, *Ice* sea ice coverage, *SSS* sea surface salinity. *T*, *u*, *v* and *q* are three dimensional fields. Best CMIP3 is the model with the best total performance in CMIP3 according to this metric, similarly worst CMIP3 is the model with the worst total performance

Xie and Arkin (1996). For the ocean, we use the World Ocean Atlas 2001 (Conkright et al. 2002) and for sea ice we use data from HadISST1 (Rayner et al. 2003). For some analysis the sea surface temperature data of Kaplan et al. (1998) has been used. We also compare the results to the CMIP3 climate model ensemble (Meehl et al. 2007).

4 Model evaluation

4.1 Atmosphere, present-day conditions

For any climate model to be stable, the top of the atmosphere radiation components should balance and also the surface net heat and freshwater flux should be zero. EC-Earth displays a

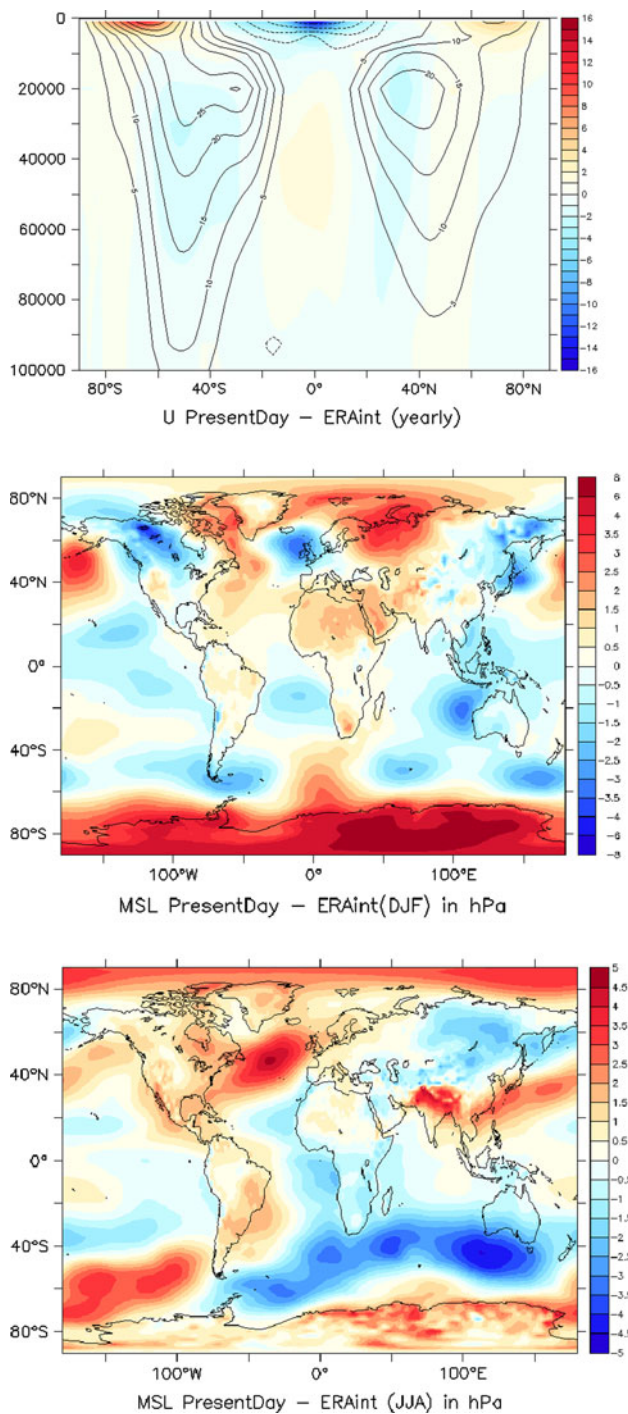


Fig. 3 Top Zonal mean zonal wind (m s^{-1}) as function of pressure level (Pa) in the EC-Earth PD simulation (contours) and the difference with ERA-Interim data (colours). Difference in sea level pressure between EC-Earth PD and ERA-Interim data (in hPa) in December, January, February (middle) and June, July, August (bottom)

near zero balance at the top of the atmosphere (TOA) and at the surface. The net top of the atmosphere radiation in the PD simulation is -0.1 W m^{-2} and the net surface energy budget is 0.6 W m^{-2} . The different shortwave and longwave

contributions are compared in Table 1 to observational estimates of Kiehl and Trenberth (1997), Trenberth et al. (2009), and ERA-Interim data. EC-Earth's radiation budget is within the uncertainties of those estimates. The net solar radiation at the TOA is too large compared to estimates of Trenberth et al., probably due to a too low cloud cover and consequently too low planetary albedo. The excess incoming shortwave radiation is balanced by the outgoing longwave radiation, resulting in a nearly closed TOA radiation budget. As with any comprehensive climate model, a small imbalance at the top of the atmosphere and surface remains associated with numerical errors. The small imbalances lead to a small drift in the control climate, well displayed in the heat and freshwater content of the ocean model (see Sterl et al., submitted to this special issue).

The meridional variation in the radiation balance implies a poleward energy transport. The atmospheric, oceanic and total energy transport, as determined from surface and TOA energy fluxes, is shown in Fig. 1. The poleward energy transport in EC-Earth is dominated by atmospheric energy transport, except in the tropics. The amplitude and the meridional variation of the energy transport resemble those derived from different reanalysis datasets by Trenberth and Caron (2001) and Fasullo and Trenberth (2008).

Next we discuss the troposphere properties of the simulated climate. The zonally averaged temperature shows a cold bias in most of the troposphere (Fig. 2). The bias is smaller when compared to ERA-40 (not shown), showing uncertainties in the reanalyses and warming of twentieth century climate. However, 2-m temperatures show that EC-Earth is too cold in the tropics and over land in boreal summer, regardless of which data set is used (Fig. 2). A strong warm bias is found in the Southern Hemisphere extratropics close to the surface. The bias is most widespread in austral summer. A localized strong bias is found in the austral winter in the South Atlantic sector near the sea ice edge, where EC-Earth displays a too small coverage of sea ice and where the oceanic mixed layers are thin. The warm Southern Ocean bias is common to coupled models that use NEMO for the ocean (e.g. Salas y Méria et al. 2005). Although part of ongoing work, there are indications that it is related to mixed layer parameterizations in NEMO. There are also indications that it is related to misrepresentation of low clouds, leading to too much shortwave radiation reaching the surface.

Although the tropics are too cold in general, the stratocumulus areas west of the continents in the subtropics stand out as warm areas. This is the result of an underestimation of the coverage of low clouds by the model. Nevertheless, the overall distribution of tropospheric temperature and humidity is well simulated (see summary in Table 2), which could be the result of the sophisticated parameterizations that are applied in an NWP model like IFS. The specific humidity

bias has a similar structure as that for temperature with a too low specific humidity in tropical regions compared to ERA-Interim data (not shown).

Over the continents, the 2-m temperature displays a similar pattern as the tropospheric biases. However, a pronounced seasonal variability in biases over the continents exists. Initially, EC-Earth displayed up to 4 K too high temperatures over Siberia and Canada in winter. The new snow parameterizations that are included in EC-Earth (Dutra et al. 2010) reduced this initial bias of IFS cycle 31r1 by almost a factor of two. The annual mean bias is now within 1 K over most continents. In boreal winter still a warm bias is found in the northern midlatitudes. In boreal summer, nearly all land area is too cold.

Strong subtropical jets are clearly present, in thermal wind balance with the temperature distribution (Fig. 3). The jet cores are very well simulated in the Northern Hemisphere. At the jet core a small underestimation of the wind speed is found. This is consistent with the weak meridional temperature gradient. The wind stress at the surface, however, is well simulated. As a consequence, the Ekman transport in the Southern Ocean has a realistic strength. The temperature bias mentioned above is therefore likely to be generated by ocean physics or atmospheric radiative fluxes, rather than the winds.

The annual mean sea level pressure deviations from ERA-Interim are generally smaller than 2 hPa. Seasonal variations in the bias show somewhat larger deviations from observations (Fig. 3). This is very small compared to other models (see summary at the end of this section). Most other coupled

models use a coarser resolution compared to EC-Earth, but sensitivity studies indicate that that convection parameterization has more impact than horizontal resolution. The only major bias is found in Antarctica where the pressure is too high, primarily due to a too high surface pressure in austral summer. A persistent bias of the IFS cycle 31r1 atmosphere model and previous cycles of the ECMWF model is the relatively high surface pressure along 60°S in the Pacific sector. In the Northern Hemisphere errors are very small. Compared to the standard IFS cycle31r1 atmosphere model, biases are reduced by more than a factor of two in the mid-latitudes of the Northern Hemisphere, mostly as a result of the new deep convection algorithm (not shown).

The model representation of stationary eddies is shown in Fig. 4. We focus on the Northern Hemisphere because of the strong zonal asymmetries induced by topography and land sea contrast. The stationary wave pattern has most structure in the winter season and is of importance in shaping the circulation that transports heat and moisture poleward. The stationary eddy component of geopotential height at 500 hPa shows that the ridge over the North Atlantic and Western Europe is displaced somewhat to the east compared to ERA-Interim. Also, the trough over East Asia and the western Pacific is slightly displaced. The strongest bias is present in the Pacific sector. This positive bias relates to surface pressure biases of about 3 hPa in the Pacific sector in boreal winter, which is small compared to other coupled models (e.g. van Ulden and van Oldenborgh 2006).

The precipitation simulated by the model is shown in Fig. 5 and compared with the CMAP data from Xie and

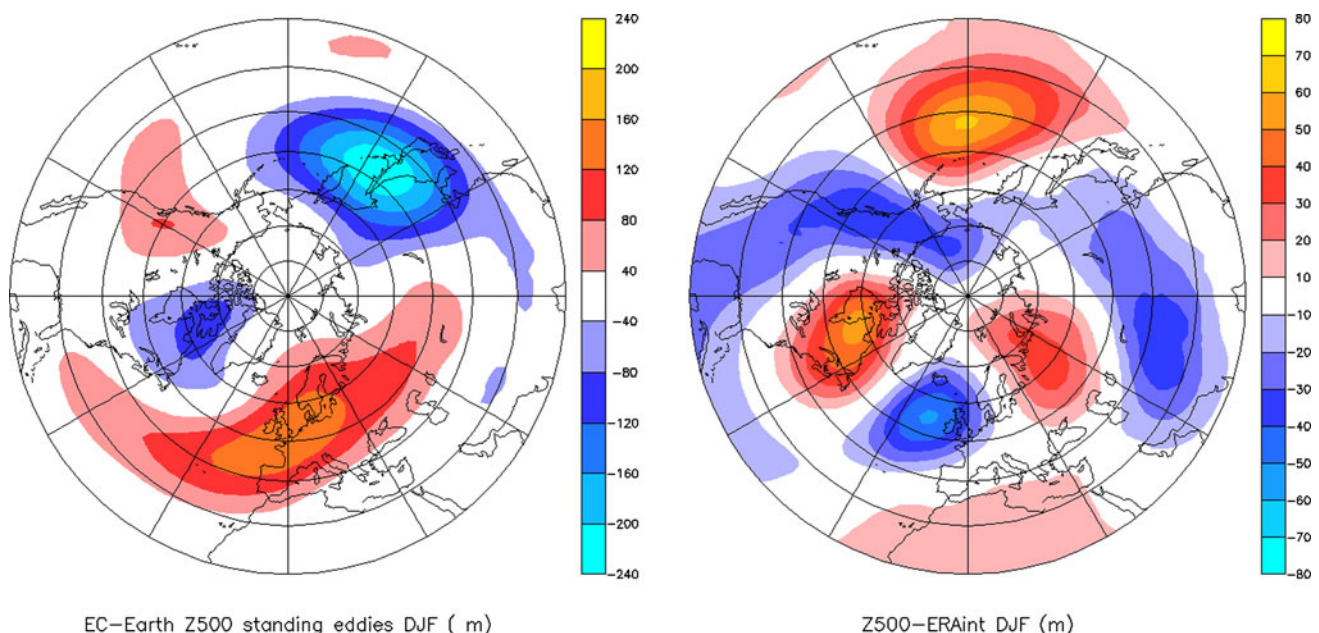


Fig. 4 *Left* Stationary eddy component (departure from zonal mean) of the 500-hPa geopotential height (m) in boreal winter in the EC-Earth PD simulation; *right* the difference with ERA-Interim. Note the difference in colour scale between the two panels

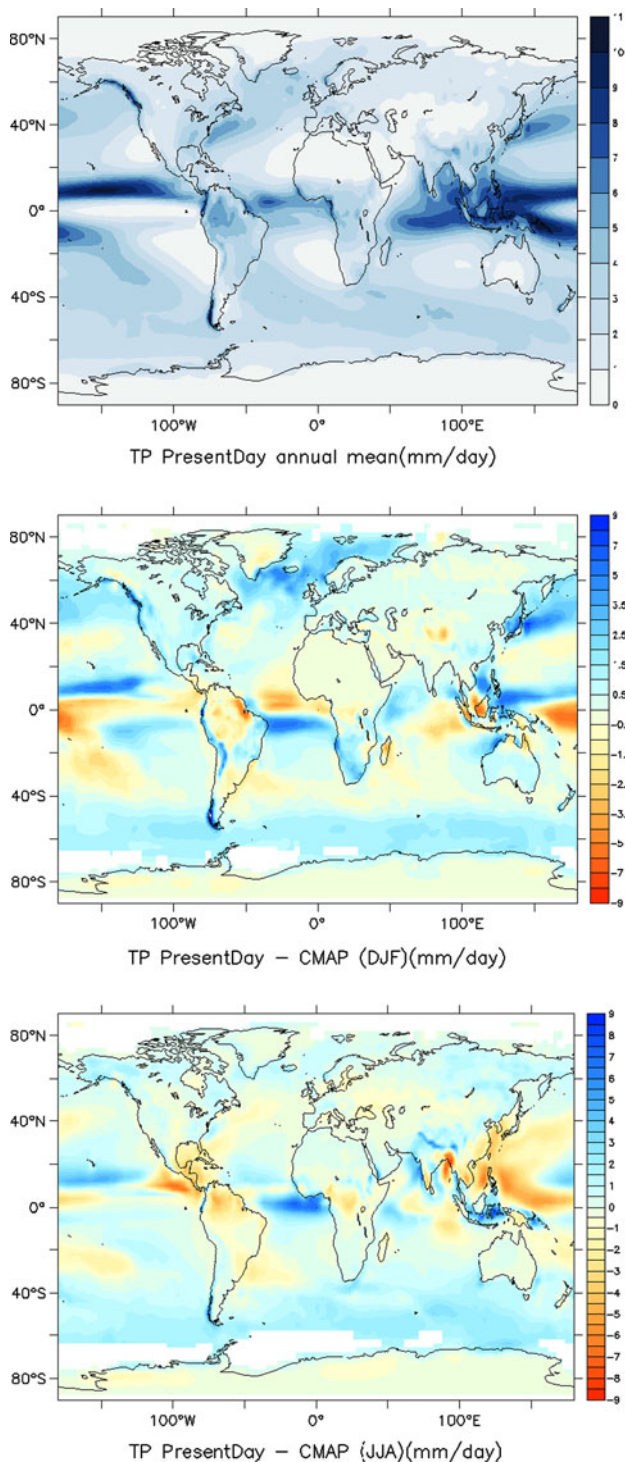


Fig. 5 *Top* Precipitation in the EC-Earth PD simulation (mm/day). Difference between the EC-Earth PD simulation and observational estimates from CMAP in December, January, February (*middle*) and June, July, August (*bottom*)

Arkin (1996). EC-Earth represents the main tropical rainbands: the Intertropical Convergence Zone (ITCZ) in the Atlantic and the Pacific and the South Pacific and South Atlantic convergence zones. EC-Earth displays too little

precipitation in the Pacific ITCZ. There is some tendency towards a double ITCZ in the Pacific reflected by the South Pacific convergence that is too weak and too zonal. In the tropical South Atlantic the rainfall is too large which indicates a southward shift of the ITCZ which crosses the equator in austral summer. As a consequence of these rainfall patterns, the Indian monsoon is well simulated, but the Asian monsoon is too weak, while in West Africa there is slightly too much rainfall in the Guinea region. In the main convective regions over South America and tropical Africa a negative bias is found. In the Pacific warm pool the biases have a more complex structure. In general, there is not enough rainfall over land in the convective zones.

The previous analyses showed different aspects of the simulated mean present-day atmosphere in the fully coupled EC-Earth model. In order to make an objective comparison between climate models, Reichler and Kim (2008) developed a metric for the simulated mean climate. The metric is based on the normalized error variance (e^2) derived from the simulated and observed climatologies of different variables at every gridpoint (interpolated on the grid of the observations). The variance of the error is scaled by the observed interannual variance:

$$e^2 = \sum_n (w_n (\bar{s}_{vn} - \bar{o}_{vn})^2 / \sigma_{vn}^2)$$

Here w_n are the weights for proper area and mass averaging, s_{vn} is the simulated climatology for climate variable (v), and grid point (n); o_{vn} is the corresponding observed climatology; and σ_{vn}^2 is the interannual variance from the validating observations. The overbar indicates that we use monthly climatologies.

This metric has been calculated for EC-Earth as well, using the same observational data as Reichler and Kim (2008) (see their paper for details). The performance index PI combines a set of Reichler and Kim metrics into a single number (PI = 1 relates to the mean of the CMIP3 results).

Table 2 shows the results for CMIP3 and EC-Earth, essentially a summary of the results described above. As expected from the results shown above, EC-Earth stands out in representing circulation features and the vertical structure of the atmosphere. Sizeable errors are found in the 2-meter temperatures and surface properties such as the net heat fluxes (although the observations are not well constrained) and wind stresses. The Northern Hemisphere is systematically better simulated than the Southern Hemisphere (not shown).

4.2 Ocean, present-day conditions

The analysis of the ocean component of EC-Earth is described in detail by Sterl et al. (submitted to this special

Fig. 6 Seasonal cycle of the sea surface temperature (K) along the equator (5°N–5°S) depicted as a Hovmöller diagram as simulated in the EC-Earth PD run (colours) compared to the observation based estimate from ERA-40 (contours). The numbers on the vertical axis refer to the months, starting from January

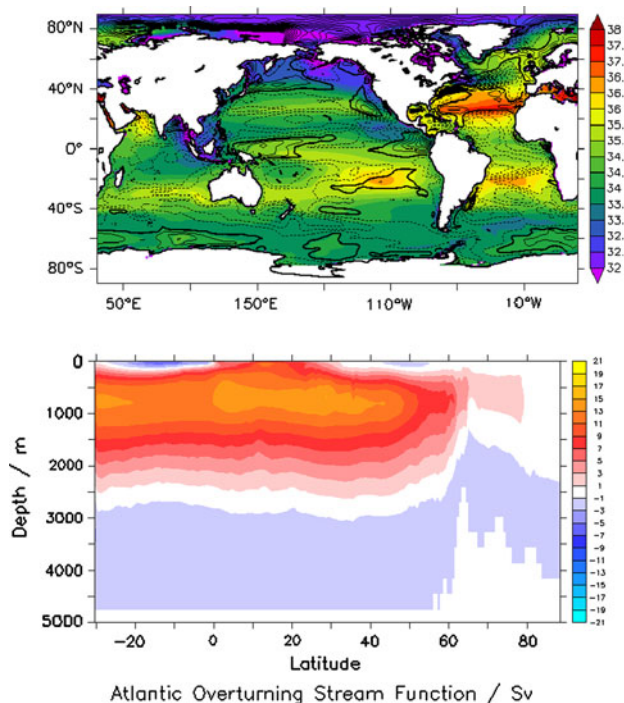
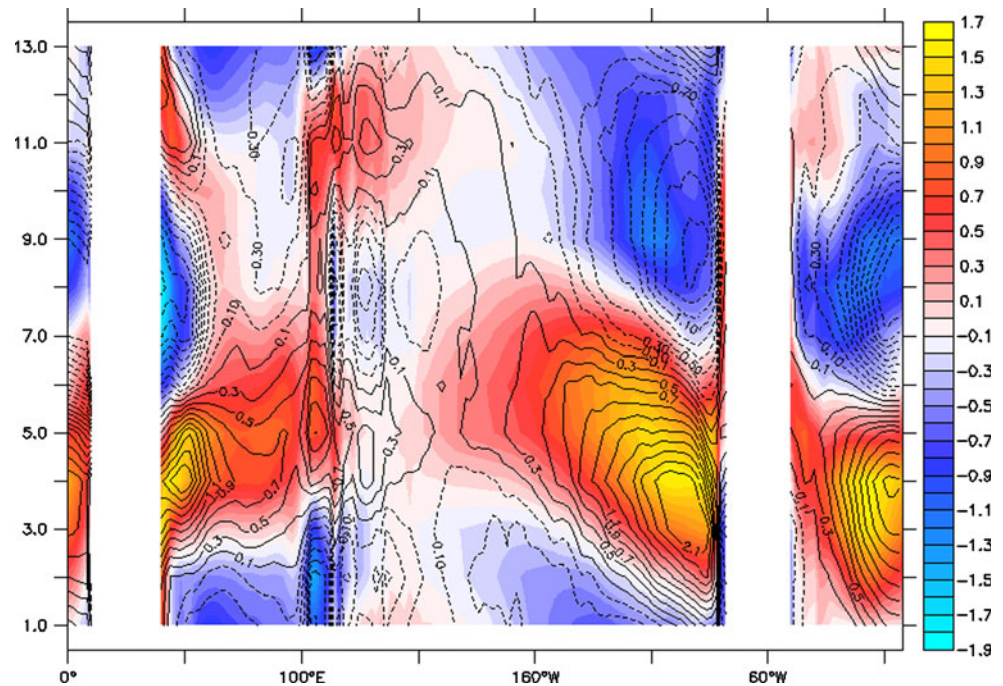


Fig. 7 *Top* annual mean sea surface salinity (psu) simulated in the EC-Earth PD run (colours) and the difference with World Ocean Atlas (contours, interval 0.2 psu). *Bottom* Atlantic meridional overturning circulation (1 Sv = $10^6 \text{ m}^3 \text{ s}^{-1}$)

issue). Here, a short assessment of the simulation of surface parameters and the Atlantic meridional overturning circulation is presented.

The sea surface temperature in EC-Earth displays similar biases as the 2-m temperature shown in Fig. 2. There is a cold bias in tropical and subtropical regions and a warm bias at higher latitudes, in particular in the Southern Ocean. Also the western boundary current extensions of the Gulf Stream close to the US coast and the Kuroshio are too warm. There is a weak cold tongue bias on the equator in the Pacific, which is favourable for simulating tropical modes of coupled variability (see Sect. 5). The cold bias has a broad scale structure and seems to originate from the atmosphere. Ocean-only runs do not display the Southern Ocean warm bias and indicate that heat fluxes, computed from bulk transfer equations with prescribed atmospheric temperature, humidity and winds, are damping the warm bias. In the coupled run, the atmosphere adjusts, the strong damping does not occur, and the bias is displayed. This indicates a contribution from ocean physics to the bias, which is likely associated with the vertical mixing in the model. The bias is most wide spread when the mixed layer restratifies in summer. This analysis is not conclusive though. As mentioned before, the error may also originate from the short wave forcing of the atmosphere model.

The temperature in the upper layers (down to 500 m) shows no significant trend in the PD run. Due to the limited length of the simulation there is drift in the deeper ocean, below 500 m the ocean warms up with about 0.03 K/century. Salinity generally decreases between 200 m and 1,500 m at typical rates of 0.02 psu/century, and does not change in deeper layers. These trends vary a lot with depth and region.

The seasonal cycle of sea surface temperature is important for the monsoons and for tropical patterns of

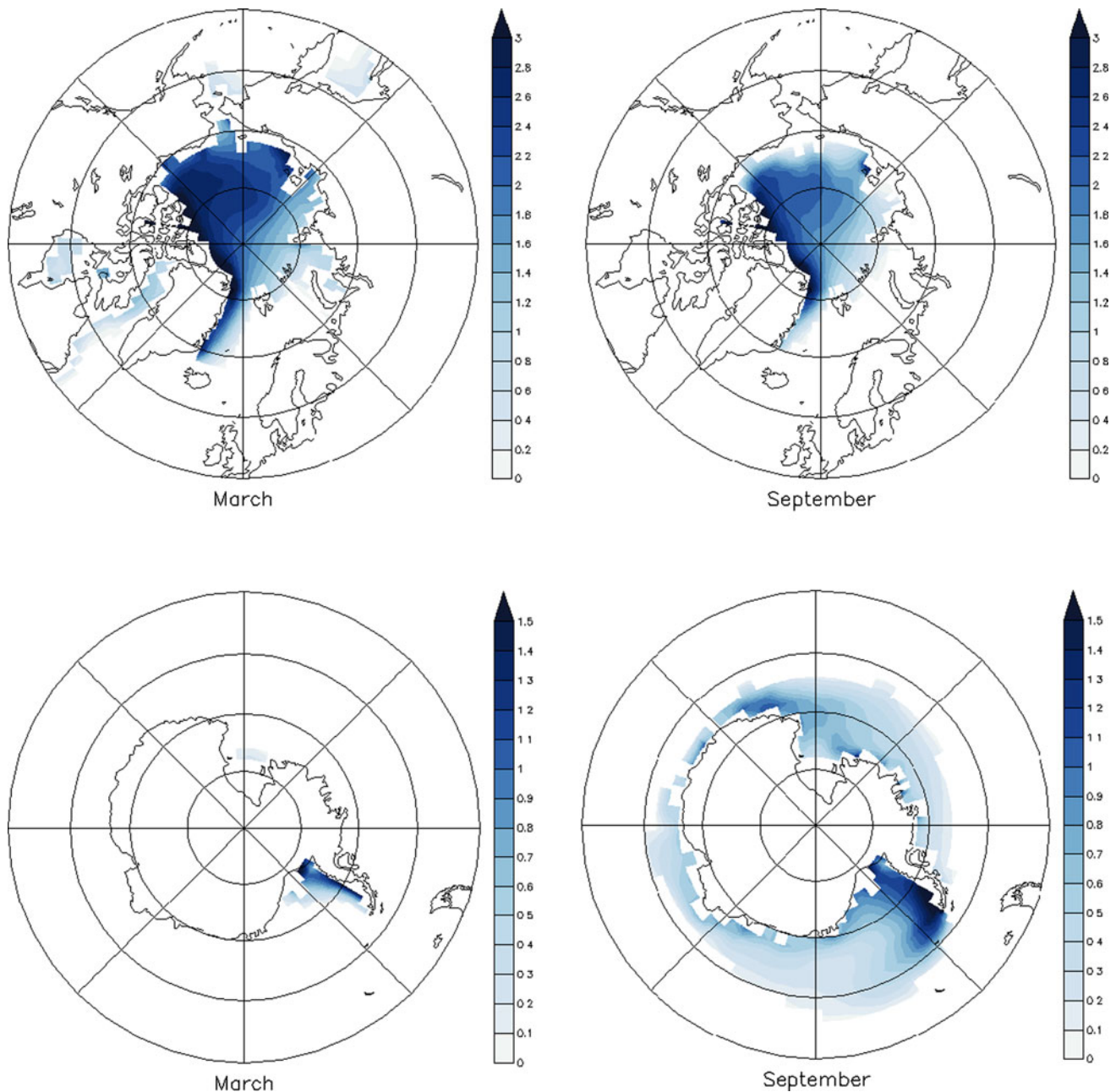


Fig. 8 March and September monthly mean sea ice thickness (m) in the Arctic (*top*) and Antarctic (*bottom*) in the EC-Earth PD simulation

variability that will be discussed later. Figure 6 shows that the major patterns of seasonal warming and cooling are well simulated in the tropics. The main bias is that the warming and cooling of the cold tongues in the Pacific and Atlantic occur too far away from the coast. This is a problem in the Atlantic in particular, where there is a weak cold tongue development that is common to many models (Hazeleger and Haarsma 2005). Also, the Pacific cold tongue develops about a month too early.

In general, sea surface salinity is too low in EC-Earth. It shows major biases in the Arctic, but the observations are not well constrained there (Fig. 7). The Indian Ocean and large parts of the Atlantic are too fresh indicating too much precipitation compared with evaporation. However, as shown in the performance metric in Table 1, the errors are of comparable magnitude as other models with similar complexity.

An important characteristic of the ocean circulation is the meridional overturning circulation. It transports large amounts of heat northward, generating a near-zero

poleward heat transport in the Southern Hemisphere (Fig. 1). In the Atlantic it is characterized by a basin wide overturning, transporting heat northward at all latitudes. The meridional overturning circulation in the Atlantic is shown in Fig. 7. The maximum overturning is located in the subtropical North Atlantic and is about 16 Sv ($1 \text{ Sv} = 10^6 \text{ m}^3 \text{ s}^{-1}$). This is weaker than estimations reported by Cunningham et al. (2007). The overturning strength displays a multidecadal oscillation with a period of about 60 years, which is discussed in detail by Wouters et al. (submitted to this special issue).

4.3 Sea ice, present-day conditions

The final component we assess is sea ice. The thickness distribution in the Arctic shows a maximum just north of Greenland (Fig. 8, top row), which is in accordance to estimates from observations, but the amplitude is too high (Belchansky et al. 2008). Also, the impact of the warm water in the northward flowing North Atlantic current is visible. The seasonal variation of thickness is slightly shifted (not shown), causing relatively thick ice in June, July and August compared to the estimates of Belchansky et al. Since sea ice thickness is not well observed, we compare the sea ice extent to observations of HadISST1 (Rayner et al. 2003). In the Arctic, the EC-Earth ice coverage match very well to the observations, with the right phasing and amplitude of the seasonal cycle (Fig. 9). It is unclear why large deviations in thickness are found, while the coverage is very well simulated.

In the Antarctic the simulated sea ice coverage is negatively biased (Figs. 8, 9). The phasing of the seasonal cycle is correct, but there is an underestimation of about 20% in amplitude, which is likely related to the Southern Ocean warm temperature bias. A more extensive analysis

of high latitude feedbacks is ongoing. The role of the stable planetary boundary layer is highlighted in a recent study by Bintanja et al. (2011) and in an extension of that study by Bintanja et al. (submitted to this special issue).

5 Patterns of interannual variability

In the previous section the mean climate and some aspects of the mean seasonal variations were presented. An important characteristic of Earth's climate is its interannual variability. Here, we present the main patterns of variability in the EC-Earth PD simulation and compare them to observations.

5.1 El Niño-Southern oscillation and pacific decadal variability

The most dominant pattern of interannual variability is the El Niño-Southern Oscillation (ENSO, see e.g. McPhaden et al. 1998, Neelin et al. 1998). Figure 10 shows linear regression patterns obtained by regressing SST onto a cold-tongue index defined in the same way as in Zhang et al. (1997). Linear regressions are performed with a 10-year high-pass and with a 15-year low-pass filter to distinguish the interannual ENSO from the Pacific Decadal Variability. The typical ENSO pattern straddling along the equator is found for the interannual variability in accordance to earlier results of EC-Earth reported by Hazeleger et al. (2010). The pattern has a stronger amplitude on the equator compared to observations. However, the observations have a coarse resolution of about 5° , while NEMO has a resolution of less than 0.5° on the equator. The pattern extends too far to the west, but has a good connection to the eastern upwelling region. The atmospheric aspects of ENSO, the winds and

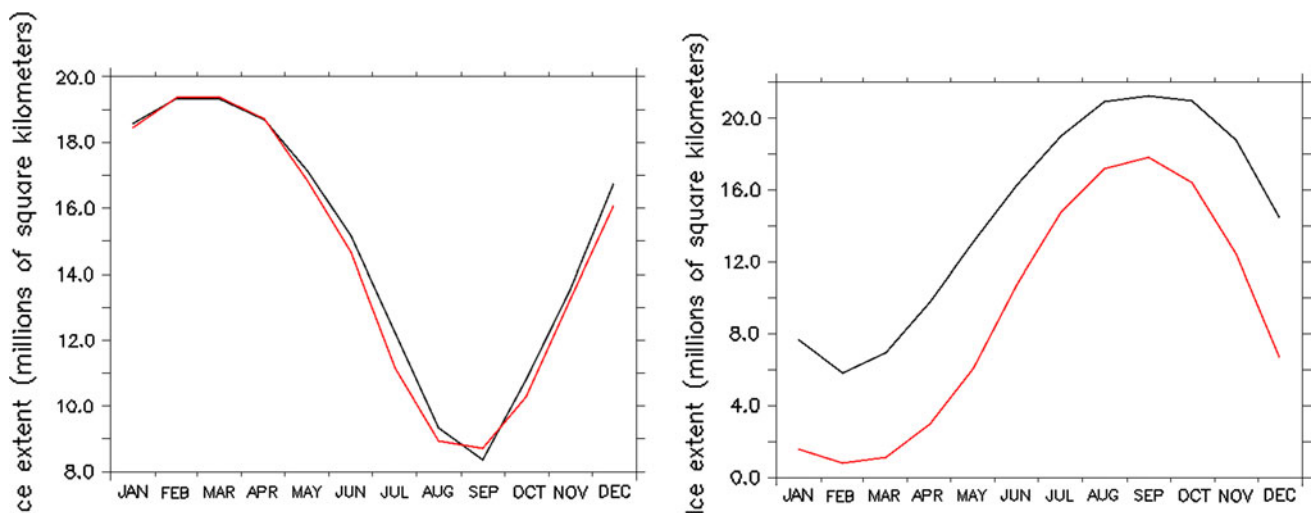


Fig. 9 Seasonal cycle of sea ice coverage (grid points with more than 15% ice coverage) in the Arctic (left) and Antarctic (right). The red line is the data from the EC-Earth PD simulation, the black line is obtained from HadISST1

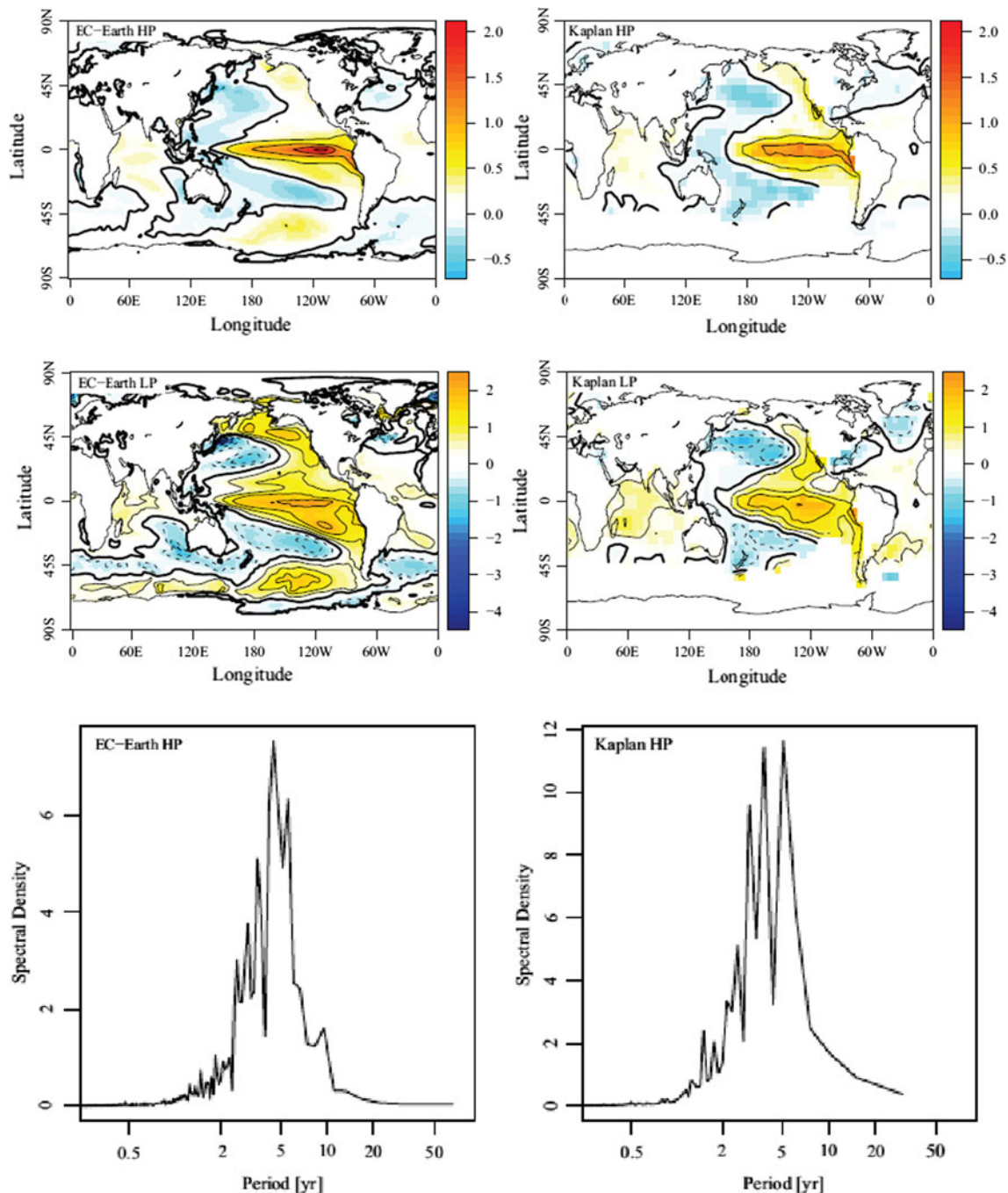


Fig. 10 *Top* Sea surface temperature anomalies regressed on a normalized cold-tongue index (5°N – 5°S , 130°E – 80°W as in Zhang et al. 1997) for 10-year high-pass filtered data from the EC-Earth PD run (*left*) and from Kaplan et al. (*right*). *Middle* As in top panel, but

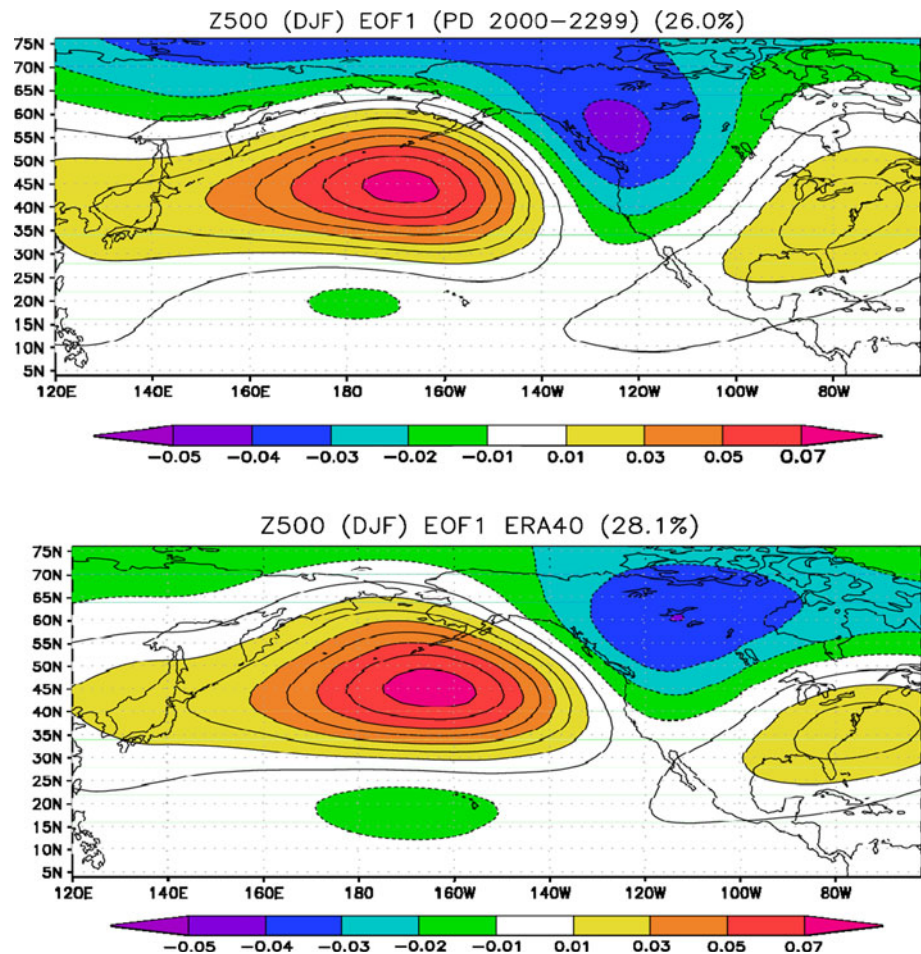
sea level pressure distribution, are well simulated too (not shown). The dominant time scale of variability can be shown in power spectra of the cold-tongue index. They show pronounced variability between 2 to 7 years in both observations and the model. However, the amplitude of ENSO is weaker than observed, the standard deviation of the cold tongue index is 0.52 K for EC-Earth and 0.75 in the

for 15-year low-pass filtered data. *Bottom* Power spectra of the 10-year high-pass filtered cold-tongue index determined from 5 equally distributed windows before averaging

observations. Additional analyses have shown that the recharge oscillator model of Jin (1997) is a good description of ENSO variability in the model (not shown).

The low-pass filtered results show that the low-frequency variability is overestimated in off-equatorial regions. Similar to the results by Zhang et al. (1997), the simulated pattern has a broader meridional scale than

Fig. 11 First empirical orthogonal function of geopotential height anomalies at 500 hPa in boreal winter in the North Pacific region in the EC-Earth PD simulation (*top*) and ERA-40 data (*bottom*). The percentage of variance explained by the pattern is indicated



ENSO and there is stronger variability in the midlatitudes. The data record is too short to distinguish a pronounced time scale of variability. In the model a broad peak between 20 and 50 years is found (not shown).

5.2 Pacific-North American pattern

In the North Pacific, variability is dominated by the Pacific-North American pattern (e.g. Wallace and Gutzler 1981). This wavetrain-like pattern is displayed well in the first empirical orthogonal function (EOF) of the geopotential height at 500 hPa. The pattern is a response to anomalous convection associated with ENSO. Figure 11 shows this pattern of variability. The amount of variance in the model is close to the observational estimate. Also, the three centres of the pattern are well located. However, the pattern stretches too far to the east over the North American continent.

5.3 North Atlantic Oscillation

In the North Atlantic region the interannual variability is dominated by the North Atlantic Oscillation (NAO; e.g. Hurrell et al. 2003). This pattern has no clear time scale of

variability. Figure 12 shows the spatial structure of variability in boreal winter: the well-known dipole displaying variations in strength of the Azores high- and Iceland low-pressure area, is well reproduced. The variance that is explained by this pattern is similar to the reanalysis data, but the NAO pattern, in particular the variation in the Azores high pressure area, is somewhat weaker in amplitude.

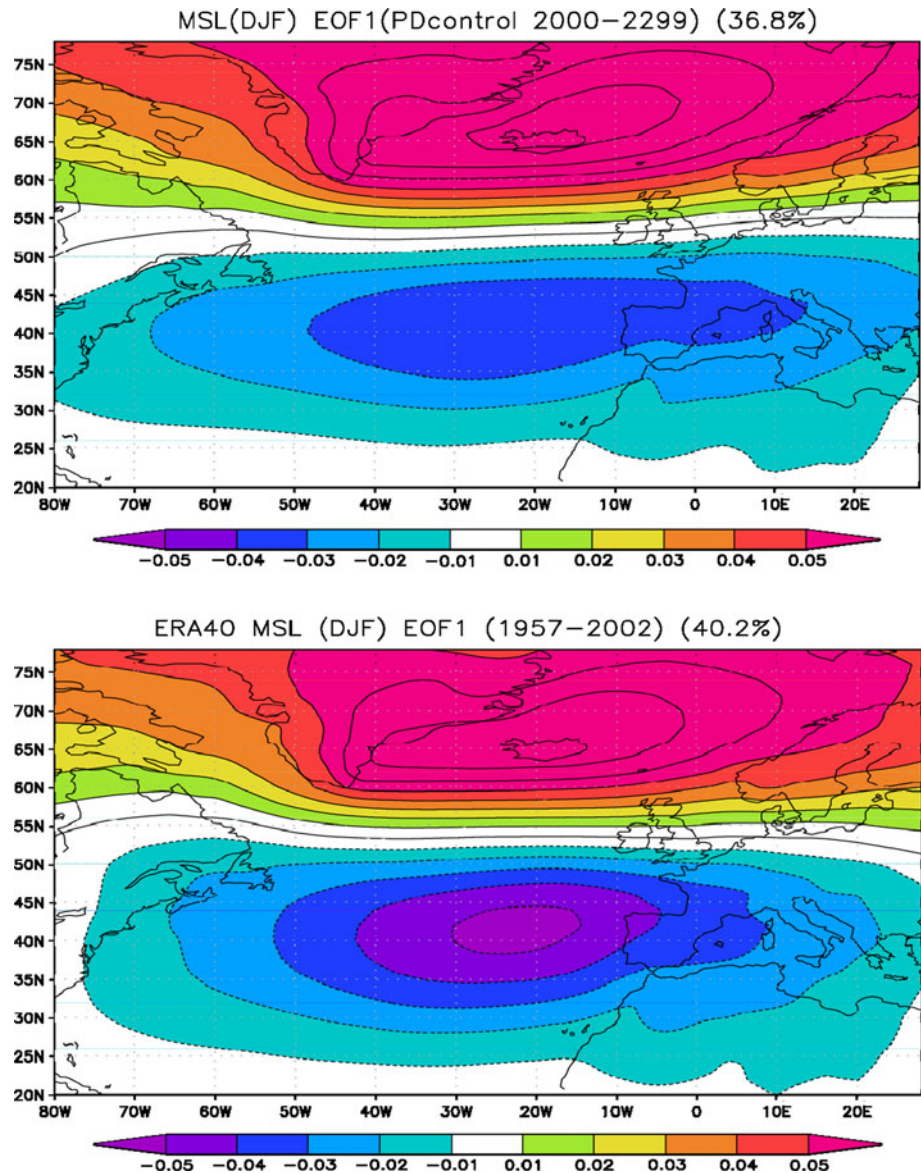
5.4 Southern Annular Mode

Finally, we show the dominant pattern of variability in the Southern Hemisphere. A zonally symmetric pattern known as the Southern Annular Mode dominates atmospheric variability (e.g. Thompson and Wallace, 2000). Figure 13 shows that this pattern is well captured by the model. The variability in the model is more symmetric than the pattern found in reanalysis data. The amount of variance explained is similar.

6 Climate sensitivities

The simulations with pre-industrial and present-day boundary conditions and with quadrupled atmospheric CO₂

Fig. 12 First empirical orthogonal function of sea level pressure anomalies in boreal winter in the North Atlantic region in the EC-Earth PD simulation (*top*) and ERA-40 data (*bottom*). The percentage of variance explained by the pattern is indicated



concentrations are used to identify the quasi-equilibrium responses to changes in greenhouse gases in the fully coupled model. A summary of the global mean responses is presented in Table 3.

The simulation with quadrupled CO_2 is too short to have reached full equilibrium. The surface temperatures hardly drift, but the net top of the atmosphere energy fluxes show that the climate still warms by 1.4 W m^{-2} . The climate sensitivity has been determined using the quasi-equilibrium method used by Winton (2006) and by a linear regression of top of the atmosphere flux against surface air temperature as shown by Gregory et al. (2004). In the latter case a sensitivity of $0.96 \text{ K}/(\text{W m}^{-2})$ is found based on the PD and $4\times\text{CO}_2$ runs, while the quasi-equilibrium method yields $0.98 \text{ K}/(\text{W m}^{-2})$. This is well within the range of uncertainty of climate sensitivity as reported in the review of Knutti and Hegerl (2008).

The warming is strongest over the Arctic (Fig. 14). There the temperature rises up to 20 K, resulting in an Arctic amplification of about 2.5 (ratio of change in temperature from 70°N – 90°N with respect to the global mean change). This is within the range of CMIP3 models of 1.5–4.5 (Holland and Bitz 2003). The continents warm more than the oceans, which is also found in CMIP3 models in warming and abrupt change simulations and which is associated with lapse rate and cloud feedbacks (e.g. Joshi et al. 2008; Laurian et al. 2009).

The zonal mean temperature change is dominated by familiar patterns of warming in the troposphere, with hot-spots in the upper tropical troposphere, in the Arctic at the surface, and cooling in the stratosphere.

The change in the zonal wind patterns is dominated by an increase of the subtropical jets in the upper troposphere (Fig. 15). This is the result of the strong stratospheric

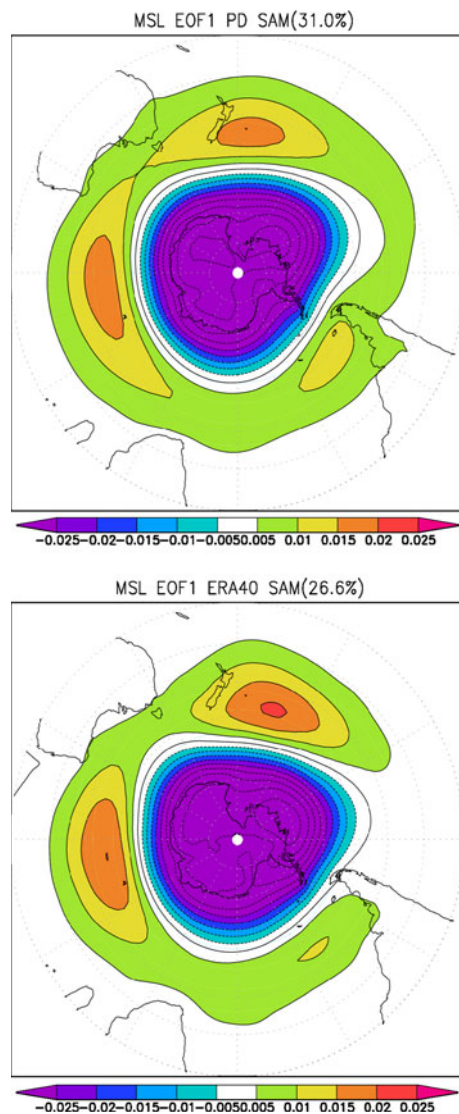


Fig. 13 First empirical orthogonal function of monthly sea level pressure variations in the Southern Hemisphere in the EC-Earth PD simulation (top) and ERA-40 data (bottom). The percentage of variance explained by the pattern is indicated

cooling and tropospheric warming. In the subtropics, where the troposphere has a strong zonal mean meridional gradient, this implies an increase in the horizontal temperature gradient and therefore an increase in zonal winds.

As can be expected the total precipitation increases in a warmer climate from 2.87 to 3.16 mm/day, which is about 1.8%/K. This is less than the 7%/K rise expected from the Clausius-Clapeyron equation, but consistent with earlier model results by e.g. Held and Soden (2006). The response of the hydrological cycle is nonlinear with respect to the change in temperature. The increase of precipitation from pre-industrial to present-day conditions is 1.1%/K. The precipitation increases in the deep tropics and high latitudes, whereas precipitation decreases in subtropical areas.

Table 3 Global mean response to changing atmospheric CO₂ concentrations

Global time mean	Pre-industrial	Present-day	4xCO ₂
2-meter temperature (K)	12.75	14.1	19.75
Total precipitation (mm day ⁻¹)	2.83	2.87	3.16
Surface shortwave radiation (W m ⁻²)	165.0	163.1	160.0
Surface longwave radiation (W m ⁻²)	−63.75	−61.5	−50.87
Surface sensible heat flux (W m ⁻²)	−19.24	−18.3	−15.9
Surface latent heat flux (W m ⁻²)	−81.53	−82.7	−90.94
Surface net energy flux (W m ⁻²)	0.48	0.6	2.29
TOA shortwave radiation (W m ⁻²)	241.3	242.6	246.6
TOA longwave radiation (W m ⁻²)	−241.5	−242.7	−245.2
TOA net energy flux (W m ⁻²)	−0.2	−0.1	1.4

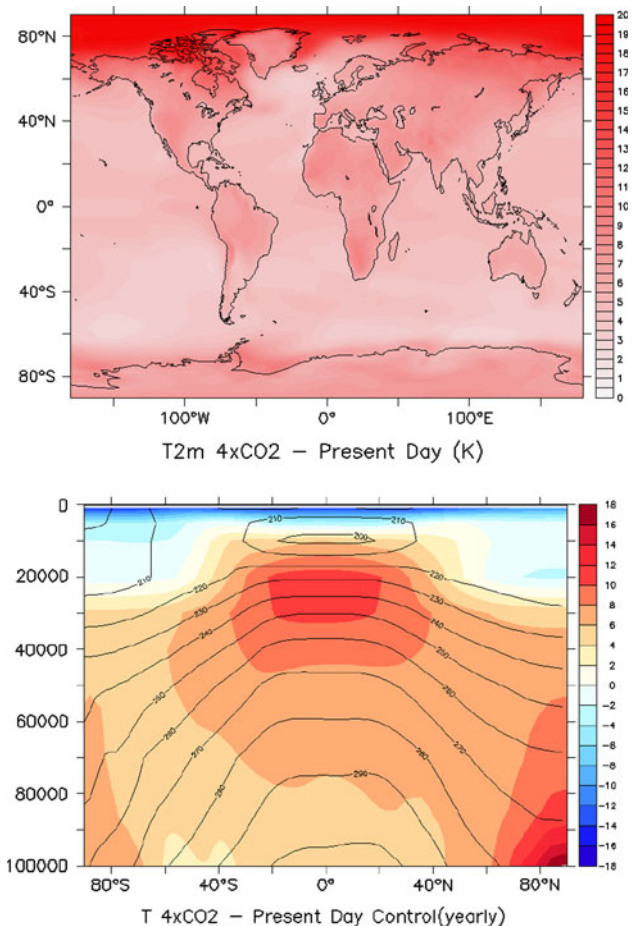


Fig. 14 Top Difference between annual mean 2-m temperature in the EC-Earth 4xCO₂ simulation and the EC-Earth PD simulation (in K). Bottom Zonal mean difference in temperature (colours, K) and the EC-Earth 4xCO₂ zonal mean temperature (contours; vertical axis in Pa)

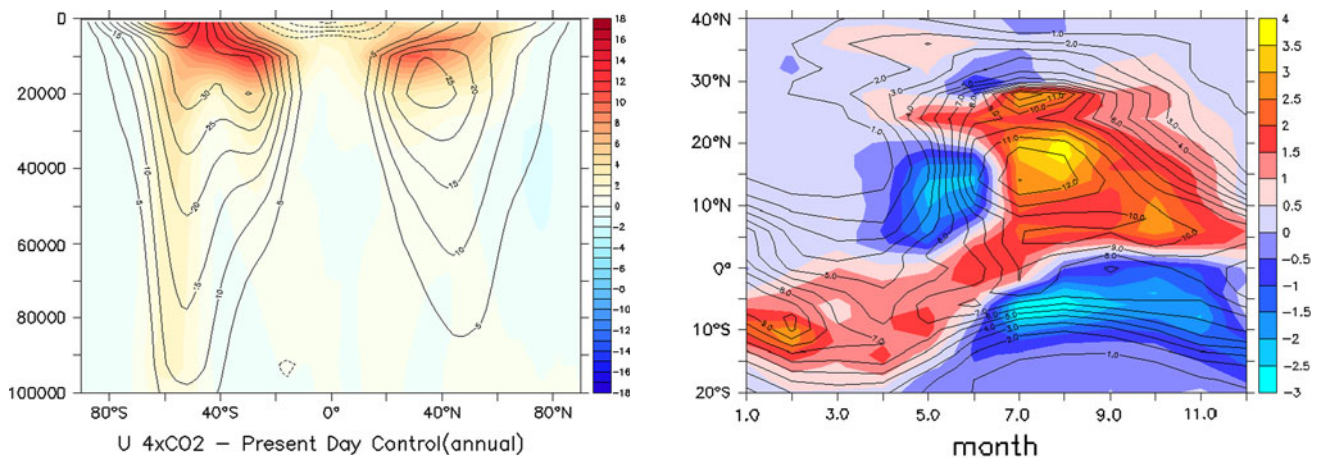


Fig. 15 Annual mean zonal mean winds (m s^{-1}) in the EC-Earth $4\times\text{CO}_2$ simulation (contours) and the difference with the EC-Earth PD simulation (colours; vertical axis in Pa)

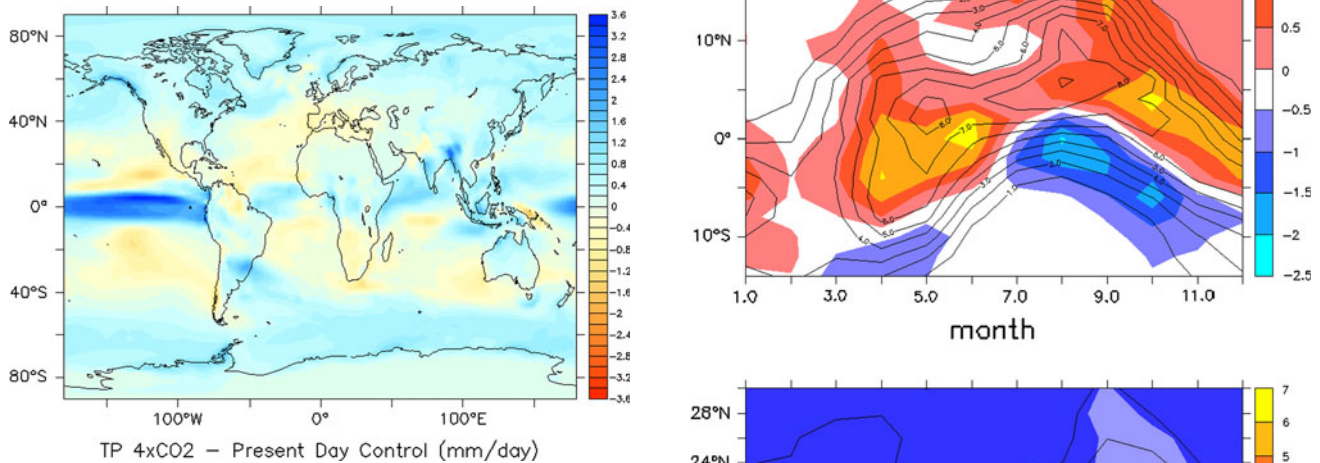


Fig. 16 Difference in annual mean total precipitation (mm/day) between the EC-Earth $4\times\text{CO}_2$ simulation and the EC-Earth PD simulation

Notably a decrease in precipitation occurs in the Mediterranean region, southwestern North America, and Southern Africa (Fig. 16).

These changes are reflected in the monsoon circulations and precipitation as well. In most cases the precipitation intensifies, but there are pronounced shifts in seasonality of the onset and demise of monsoons (Fig. 17). For instance, the Indian monsoon onset occurs later when greenhouse gasses increase. The West African monsoon shifts northward, likely to be caused by strong heating over the northern African continent with respect to the Gulf of Guinea, and it intensifies at the onset and demise of the monsoon. For North America, the change in monsoon characteristics is much larger. The drying of the southwestern part of the United States of America is clearly visible and affects the onset of the monsoon. An increase of precipitation is found in the Gulf of Mexico area and at the

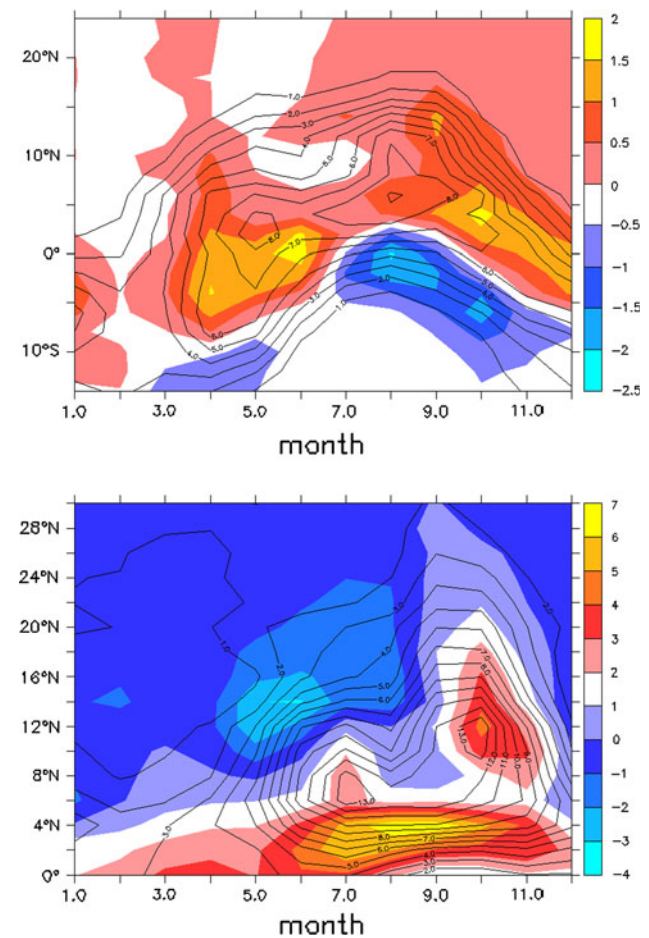


Fig. 17 Monsoon precipitation in PD (contours) and $4\times\text{CO}_2$ minus PD (colours) conditions in mm/day . *Top* Indian monsoon defined by the zonal average between 70°E and 100°E . *Middle* West African monsoon defined by the zonal average between 10°W and 20°E . *Bottom* North American monsoon defined by the zonal average between 95°W and 110°W

end of the monsoon season. These non-trivial changes in monsoon characteristics require further study, which is beyond the scope of this paper.

The patterns of interannual variability hardly change between the three different runs. However, the temporal variability of ENSO shows a very marked change. ENSO is very similar in the PD and PI runs, but becomes extremely regular with a period of somewhat less than three and a half years in the $4\times\text{CO}_2$ simulation. In the PI and PD run it appeared that although ENSO is irregular it shows more regularity than the observations. It is less skewed and in particular the transition to the cold phase of ENSO is more regular. When CO_2 is quadrupled with respect to the PD simulation ENSO seems to become a self sustained oscillator in EC-Earth.

7 Conclusions

In this paper the new Earth system model EC-Earth version 2.2 is presented, based on the seasonal forecasting system 3 of ECMWF (System 3). Few modifications to the atmosphere model of ECMWF were made and a different ocean model and land-surface model have been coupled to the atmosphere. These new modules are likely to be included in future seasonal forecasting systems that are being developed at ECMWF. EC-Earth will be used for transient climate integrations as part of the CMIP5 project. In that case different greenhouse gas concentrations, aerosol loadings and land use fields will be used. Furthermore, the aerosol optical properties are somewhat different.

The large-scale characteristics of the atmosphere, such as the distributions of sea level pressure, temperature and humidity distributions are well simulated in a long present-day control run. When compared to models of similar complexity biases of similar magnitude are found. The largest biases are found in the surface parameters. On average, the modelled atmosphere is too cold, except over the Southern Ocean and Northern Hemisphere extratropics. EC-Earth performs well in atmospheric circulation characteristics as displayed in the small bias in sea level pressure.

The model features realistic interannual variability. Generally, the spatio-temporal characteristics are well simulated and patterns explain similar amounts of variance as in reanalysis data. Also, for some patterns of variability (ENSO) the mechanisms of variability resemble observations closely. These aspects seem to be better simulated than in many CMIP3 models. It should be noted that these models were generally run at lower spatial resolution.

Experiments with changing CO_2 concentrations in the atmosphere show similar responses as found in earlier model studies: the tropospheric warming with a maximum in the higher tropical troposphere, stratospheric cooling and polar amplification. Changes to the hydrological cycle also show similar results. The model's climate sensitivity is

within ranges found in the literature. Changes to the hydrological cycle are comparable to that found in other studies with an intensification as greenhouse gases rise and a weaker increase in precipitation than in humidity. However, this change is nonlinear with respect to change in temperature, with stronger increases at stronger temperature increases. In general the increased hydrological cycle causes intensifications of monsoon precipitation, with the exception of the North American monsoon, which features a reduction at the onset due to the strong drying in the subtropics. Interannual variability hardly changes, except for ENSO which becomes a regular oscillation when CO_2 is quadrupled.

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